A PROJECT REPORT

ON

Geo-Tracker App

A Course Project report Submitted in partial fulfillment of the Academic requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGNECE AND MACHINE LEARNING

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SRI INDU COLLEGE OF ENGINEERING & TECHNOLOGY(Autonomous)

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2025-2026

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CERTIFICATE

This is to certify that the Project entitled **"Geo-Tracker App"** is submitted in partial fulfilment for the award of the **Degree of Bachelor of Technology in Artificial Intelligence and Machine Learning** of Sri Indu College of Engineering and Technology, Hyderabad for the academic year 2025–2026. This is a record of the project work done by **S. Shirisha (23D41A66G9), T. Harini (23D41A66H0), T. Rishwanth (23D41A66H1), U. Navya (23D41A66H2), and U. Akash (23D41A66H3),** which has been approved as it satisfies the academic requirements in respect of the work prescribed for **III Year, I-Semester of the B.Tech course.**

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ABSTRACT

The **Geo-tracker App** is a simple Flutter-based mobile application that retrieves the user's current geographical location using GPS services. The project demonstrates the integration of device sensors with Flutter applications through the use of the **Geolocator** and **Geocoding** packages. When the user taps a button, the application fetches the device's latitude and longitude coordinates and converts them into a readable address including city, state, and country.

This project highlights key concepts such as permission handling, accessing device hardware, asynchronous programming, and real-time data display on the user interface. The application is lightweight, fast, and designed to run on any Android device or emulator, making it ideal for learning basic mobile development with Flutter. The system can be extended in future to include Google Maps integration, live tracking, navigation features, or location sharing, making it a strong foundation for more advanced location-based applications.

INTRODUCTION

The **Geo-Tracker App** is a lightweight, cross-platform mobile application designed to determine and display a user's real-time geographic location using GPS technology. In today's digital environment, location-based services have become essential components of mobile applications. From navigation systems and delivery tracking to emergency assistance and personalized recommendations, accurate location detection plays a crucial role in enhancing user experience and enabling smart mobile solutions.

Flutter, Google's UI toolkit for building natively compiled applications, provides an ideal development platform for creating such location-aware apps due to its performance, portability, and ease of use. The Geo-Tracker App utilizes two major Flutter packages—**Geolocator** and **Geocoding**—to gather GPS coordinates and convert them into meaningful, human-readable address information. Geolocator offers a powerful set of APIs for retrieving latitude, longitude, and permission status, while Geocoding enables the translation of raw coordinates into structured address components such as street name, city, state, and country.

The application is designed with simplicity and efficiency in mind. It features a minimalistic user interface that displays location data instantly upon user request. Internally, the app handles critical aspects such as checking for GPS availability, requesting runtime permissions, managing asynchronous operations, and displaying accurate results without delays. This ensures that users receive clear and reliable information, even with limited technical knowledge.

The Geo-Tracker App not only demonstrates fundamental concepts of mobile system design—such as sensor integration, asynchronous programming, and API communication—but also reflects real-world application scenarios. Students can build upon this foundation to create advanced systems like navigation tools, emergency SOS apps, geofencing solutions, or delivery tracking systems.

By combining the flexibility of Flutter with the powerful capabilities of modern smartphones, the Geo-Tracker App highlights how simple code can drive meaningful functionality. It provides an excellent learning experience for students and a practical demonstration of how GPS-based services can be integrated into mobile applications with minimal complexity and maximum usability.

EXISTING SYSTEM

Most mobile devices today come with built-in GPS used by applications like Google Maps, Uber, Swiggy, and Zomato. These apps rely heavily on location data but are primarily large-scale commercial systems with complex features such as maps, routing algorithms, real-time tracking, and integrated databases.

For learners or small academic projects, these systems have several limitations:

1. **Lack of Simplicity:** They provide complete navigation services but hide the core workings of GPS coordinate retrieval and conversion to addresses.
2. **No Transparency:** Users cannot easily see underlying processes like permission handling, real-time GPS access, or error handling when GPS is off.
3. **Heavy Systems:** Advanced features like live maps, background tracking, and server-side APIs make them resource-intensive and complex.
4. **Limited Customization:** Students cannot easily modify these apps to fetch only coordinates, display addresses, or experiment with minimal UI.
5. **Internet Dependence:** Many apps require continuous internet for maps and syncing, though basic GPS functionality can work offline.

PROPOSED SYSTEM

The proposed system introduces a simple and efficient mobile application called **Geo-Tracker**, designed to accurately determine the user's current location. The system focuses on essential GPS functionality and avoids unnecessary complexity. The key points of the proposed system are as follows:

- **Real-Time Location Retrieval:**

The system fetches the user's current latitude and longitude using the device's GPS through the Geolocator package.

- **Address Conversion:**

The raw GPS coordinates are converted into a readable address format using the Geocoding package.

- **Simple User Interface:**

A minimal UI with a single button allows users to easily fetch and view their location details.

- **Automatic Permission Handling:**

The app checks for location permissions and prompts the user to enable them or turn on GPS if required.

- **Cross-Platform & Lightweight:**

Developed using Flutter, the app runs on both Android and iOS, providing a lightweight and efficient location tracking solution.

HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS:

Processor :- Intel Core i5 (1st Generation or above)/ Mac M-series Chip

Hard Disk :- 256 GB

RAM :- 8GB(minimum)

SOFTWARE REQUIREMENTS:

Operating System :- Windows /MAC OS

Programming Language :- Dart

Software / IDE used :- Flutter, Android Studio

IMPLEMENTATION

main.dart

```
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';
import 'package:geocoding/geocoding.dart';

void main() {
  runApp(const LocationApp());
}

class LocationApp extends StatelessWidget {
  const LocationApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: HomePage(),
    );
  }
}

class HomePage extends StatefulWidget {
  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  String locationText = "Tap the button to get your location";

  Future<void> fetchLocation() async {
    bool enabled = await Geolocator.isLocationServiceEnabled();
    if (!enabled) {
      setState(() => locationText = "Turn ON location services");
      return;
    }

    LocationPermission permission = await Geolocator.requestPermission();
    if (permission == LocationPermission.denied ||
        permission == LocationPermission.deniedForever) {
      setState(() => locationText = "Location permission denied");
      return;
    }

    Position pos = await Geolocator.getCurrentPosition(
      desiredAccuracy: LocationAccuracy.high,
    );
```

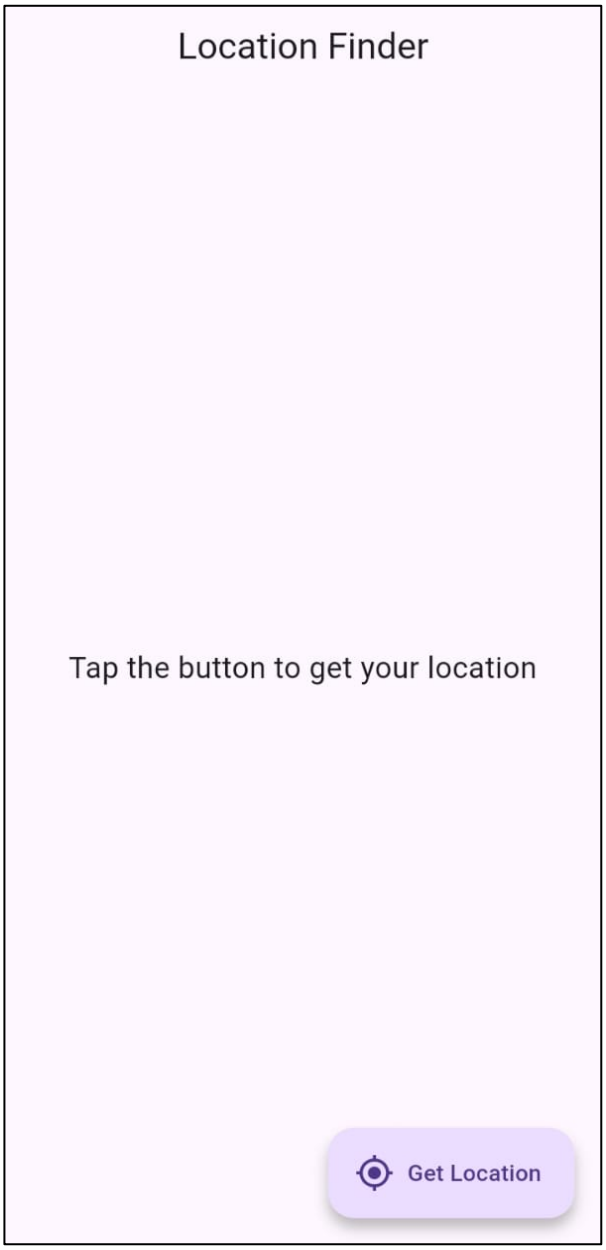
```
    List<Placemark> place = await placemarkFromCoordinates(
      pos.latitude,
      pos.longitude,
    );

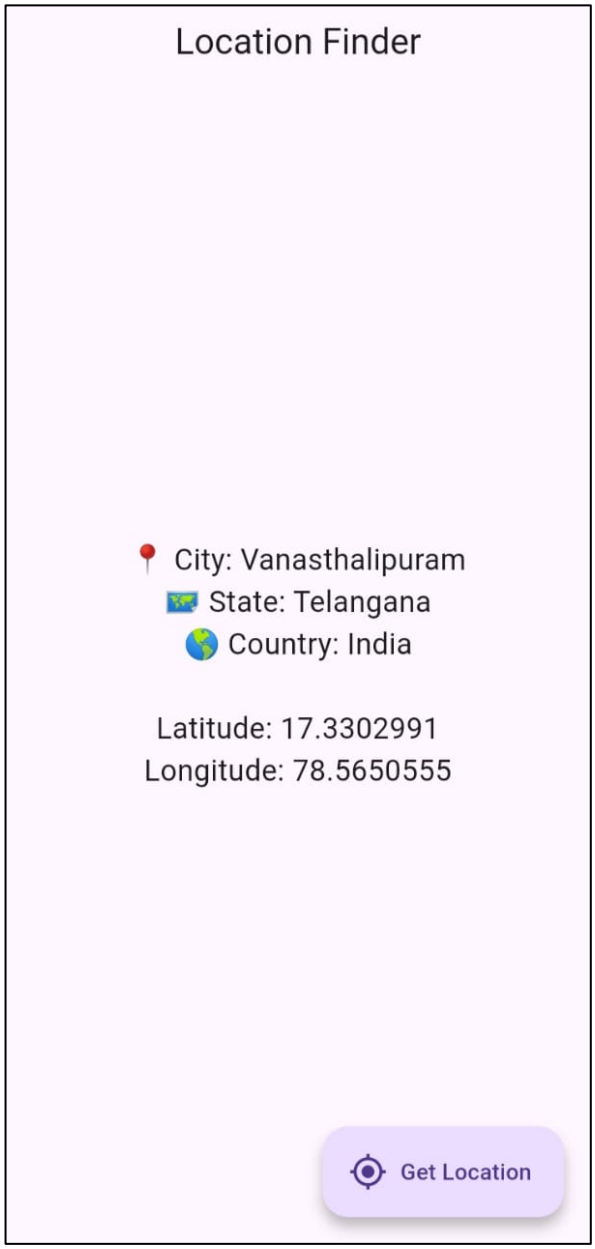
    String city = place[0].locality ?? "Unknown city";
    String state = place[0].administrativeArea ?? "";
    String country = place[0].country ?? "";

    setState(() {
      locationText =
      "📍 City: $city\n🗺️ State: $state\n🌎 Country: $country\n\n"
          "Latitude: ${pos.latitude}\nLongitude: ${pos.longitude}";
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Location Finder"),
        centerTitle: true,
      ),
      body: Center(
        child: Text(
          locationText,
          textAlign: TextAlign.center,
          style: const TextStyle(fontSize: 18),
        ),
      ),
      floatingActionButton: FloatingActionButton.extended(
        onPressed: fetchLocation,
        icon: const Icon(Icons.my_location),
        label: const Text("Get Location"),
      ),
    );
  }
}
```

RESULT


Location Finder
Tap the button to get your location
Get Location


Location Finder
City: Vanasthalipuram
State: Telangana
Country: India
Latitude: 17.3302991
Longitude: 78.5650555
Get Location

CONCLUSION

The Geo-Tracker app successfully demonstrates the core concepts of location-based services in a simple and understandable way, making it ideal for academic and learning purposes. By integrating GPS functionalities using Flutter, the app allows users to fetch real-time latitude and longitude, convert them into human-readable addresses, and display essential location details such as city, state, and country. The system handles permissions effectively, ensures error handling when GPS is disabled, and highlights the differences between offline and online location fetching.

Unlike commercial applications like Google Maps or Uber, which are feature-heavy and complex, this app is lightweight, transparent, and easily customizable. Students and developers can experiment with minimal UI, permission prompts, and core GPS functionalities without dealing with the overhead of server-side APIs, live maps, or background tracking. The simplicity of the proposed system helps in understanding how location data is obtained, processed, and displayed, providing practical insight into mobile app development.

Overall, the Geo-Tracker project emphasizes learning and experimentation while delivering a functional and user-friendly location tracking solution. It bridges the gap between theoretical concepts of GPS and their practical application in mobile development, empowering learners to build more advanced location-based applications in the future.